Mini Project 1A

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Mini Project 1A
CS 4001
Due Feb 2, 11:59pm
To Do:
Fear mechanism simulation on a local VS Code:

1. Apply a 0.2 nA current. What is the oscillation frequency? What emotional state can it correspond to?

With a 0.2 nA current, the oscillation frequency is 3.918 Hz. This oscillation frequency is low and corresponds with a no-fear / baseline emotional state where amygdala neurons spike infrequently.

```
(fear_sim) [suqdw@engr-nelvm-res fear_simulation]$ python check_output.py
The network is oscillating around 3.917727717925015 Hz.
Error flashing flicker.py to microbit: Unable to find micro:bit. Is it plugged in?
```

2. Apply a 1.0 nA current. What is the oscillation frequency? What emotional state can it correspond to?

With a 1.0 nA current, the oscillation frequency is 19.54 Hz. The emotional state that this oscillation frequency corresponds to is fear where the neurons in the amygdala are spiking rapidly.

```
(fear_sim) [suqdw@engr-nelvm-res fear_simulation]$ python check_output.py
The network is oscillating around 19.538882375937124 Hz.
Error flashing flicker.py to microbit: Unable to find micro:bit. Is it plugged in?
```

Ps. We tried the micro:bit flashing part later at home, and it worked just fine. Screenshots above were taken before that.

Computational Thinking Steps

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According to Gemini, this approach focuses on the logical implementation of the brain circuit model and the Cyberinfrastructure (CI) workflow; follows this order:

Decomposition -> Pattern recognition -> Abstraction -> Algorithmic design

Decomposition: Breaking the end-to-end CI workflow into distinct modules: the input parameter interface, the Python-based NEURON brain circuit model, and the output communication to the micro:bit LEDs.

Pattern Recognition: Identifying how different current inputs (e.g., 0.2 nA vs. 1.0 nA) consistently produce distinct oscillation frequencies that map to discrete emotional states (baseline vs fear).

Abstraction: Focuses on modeling the complex biological amygdala using a simplified NEURON software simulation which can be used to predict the respective firing rate of the amygdala.

Algorithmic Design: The created simple workflow follows:

1. The user inputs and can change the current amount

2. The model calculates the amygdala's firing rate.

3. The system translates that rate into a visual signal (low flashing vs. rapid flashing) for the micro:bit hardware.

Design Thinking Steps

This approach focuses on user interaction and the physical representation of the simulated "brain."

Empathize: Recognizing the difficulty in visualizing internal brain states and translating abstract neural activity into an intuitive, observable signal.

Define: The goal is to design a system where a user can change inputs and receive an immediate, intuitive visual prediction of a brain "state."

Ideate: The micro:bit hardware can be used to represent the amygdala data into the physical form by flashing in different frequencies.

Prototype: Making sure that the simulation works in different CI platforms, like a local PC.

Test: It is important to verify that the oscillation frequency output (such as 19.54 Hz for fear) and the flicker.py script are effectively flashing the LEDs at the appropriate relative speed.